EXPERIMENT 11: CRYPTARITHMETIC PROBLEM

AIM

To solve a cryptarithmetic puzzle by assigning distinct digits to letters so that the arithmetic equation is valid.

ALGORITHM

1. Choose a cryptarithmetic equation and identify its unique letters.
2. Assign a different digit to each letter.
3. Do not allow a leading letter of a multi-digit number to be zero.
4. Check whether the digit assignment satisfies the arithmetic equation.
5. Display a valid assignment and the resulting arithmetic equation.

PYTHON CODE

```
from itertools import permutations

# Solve SEND + MORE = MONEY
letters = "SENDMORY"
digits = range(10)

for p in permutations(digits, len(letters)):
    d = dict(zip(letters, p))
    if d["S"] == 0 or d["M"] == 0:
        continue

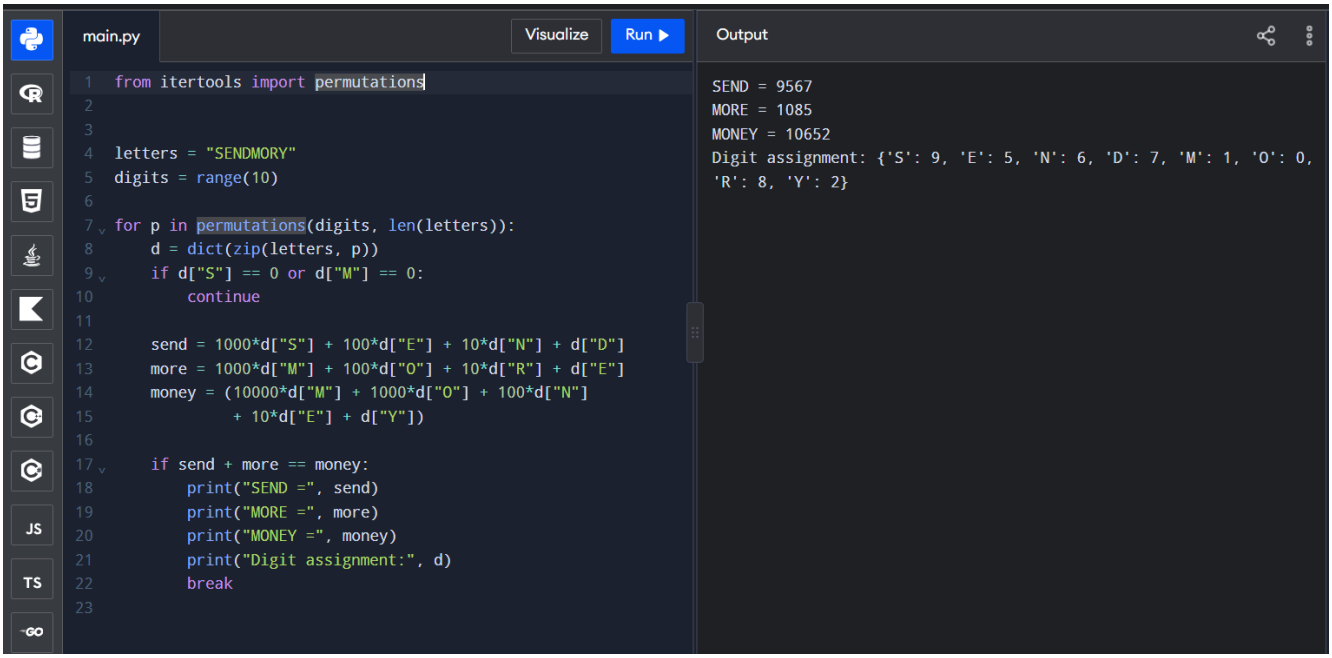
    send = 1000*d["S"] + 100*d["E"] + 10*d["N"] + d["D"]
    more = 1000*d["M"] + 100*d["O"] + 10*d["R"] + d["E"]
    money = (10000*d["M"] + 1000*d["O"] + 100*d["N"]
             + 10*d["E"] + d["Y"])

    if send + more == money:
        print("SEND =", send)
        print("MORE =", more)
        print("MONEY =", money)
        print("Digit assignment:", d)
        break
```

INPUT

```
The equation SEND + MORE = MONEY is predefined in the program.
```

OUTPUT



`SEND = 9567\nMORE = 1085\nMONEY = 10652`

RESULT

The cryptarithmetic puzzle was solved successfully by finding distinct digit assignments that satisfy the equation.